

Loops and Functions in JavaScript - Assignment Solutions

PW EARNERS : Coding and Web Development

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LEC - 31

Section 1 - Basic Loops

Part A - for Loop

1. Print Numbers

Write a program to print numbers from 1 to 10 using a for loop.

Solution:

```
for (let i = 1; i <= 10; i++) {  
    console.log(i);  
}
```

2. Print Even Numbers

Write a program to print all even numbers from 1 to 20.

Solution:

```
for (let i = 2; i <= 20; i += 2) {  
    console.log(i);  
}
```

3. Print Odd Numbers

Write a program to print all odd numbers from 1 to 20.

Solution:

```
for (let i = 1; i <= 20; i += 2) {  
    console.log(i);  
}
```

4. Reverse Counting

Write a program to print numbers from 10 to 1 using a loop.

Solution:

```
for (let i = 10; i >= 1; i--) {  
    console.log(i);  
}
```

5. Sum of Numbers

Write a program to calculate the sum of numbers from 1 to 10.

Solution:

```
let sum = 0;  
  
for (let i = 1; i <= 10; i++) {  
    sum += i;  
}
```

```
}
```

```
console.log(sum);
```

6. Multiplication Table

Take a number and print its multiplication table up to 10.

Solution:

```
let num = 5;
```

```
for (let i = 1; i <= 10; i++) {  
  console.log(`${num} x ${i} = ${num * i}`);  
}
```

Explanation: Here, 5 is the input number. You can replace it with any other number.

Part B - while Loop

7. Basic while Loop

Write a program to print numbers from 1 to 10 using a while loop.

Solution:

```
let i = 1;
```

```
while (i <= 10) {  
  console.log(i);  
  i++;  
}
```

8. Sum of Even Numbers

Write a program to calculate the sum of all even numbers from 1 to 20.

Solution:

```
let i = 2;
```

```
let sum = 0;
```

```
while (i <= 20) {  
  sum += i;  
  i += 2;  
}
```

```
console.log(sum);
```

9. Stop the Loop Using break

Write a program using a while loop to print numbers from 1 onwards, but stop the loop when the number reaches 6 using the break statement.

Expected Output: 1 2 3 4 5

Solution:

```
let i = 1;

while (i <= 10) {
  if (i === 6) {
    break;
  }

  console.log(i);
  i++;
}
```

10. Skip a Number

Print numbers from 1 to 10, but skip the number 5 using the continue statement.

Solution:

```
let i = 1;

while (i <= 10) {
  if (i === 5) {
    i++;
    continue;
  }

  console.log(i);
  i++;
}
```

Section 2 - Functions

11. Function with a Parameter

Create a function named greetUser(name) that takes a name as a parameter and displays a greeting message.

Example: Input: Rahul | Output: Hello, Rahul

Solution:

```
function greetUser(name) {  
  console.log(`Hello, ${name}`);  
}
```

```
greetUser("Rahul");
```

12. Add Two Numbers

Create a function that takes two numbers as parameters and returns their sum.

Solution:

```
function addNumbers(a, b) {  
  return a + b;  
}
```

```
let result = addNumbers(10, 20);  
console.log(result);
```

13. Even or Odd Function

Create a function that takes a number and checks whether it is even or odd.

Solution:

```
function checkEvenOdd(num) {  
  if (num % 2 === 0) {  
    console.log("Even");  
  } else {  
    console.log("Odd");  
  }  
}
```

```
checkEvenOdd(7);
```

14. Square of a Number

Create a function that takes a number and returns its square.

Solution:

```
function square(num) {  
  return num * num;  
}
```

```
console.log(square(5));
```

15. Largest of Two Numbers

Create a function that takes two numbers and returns the greater number.

Solution:

```
function largest(a, b) {  
  if (a > b) {  
    return a;  
  } else {  
    return b;  
  }  
}  
  
console.log(largest(10, 25));
```

16. Calculate Total Price

Create a function named calculateTotal(price, quantity) using a function declaration. The function should calculate and display the total price.

Example: Input: price = 100, quantity = 3 | Output: Total Price: 300

Solution:

```
function calculateTotal(price, quantity) {  
  let total = price * quantity;  
  console.log(`Total Price: ${total}`);  
}  
  
calculateTotal(100, 3);
```

Section 3 - Functions with Loops

17. Print Numbers Using a Function

Create a function printNumbers(n) that prints numbers from 1 to n using a loop.

Example: Input: 5 | Output: 1 2 3 4 5

Solution:

```
function printNumbers(n) {  
  for (let i = 1; i <= n; i++) {  
    console.log(i);  
  }  
}  
  
printNumbers(5);
```

18. Multiplication Table Function

Create a function `printTable(num)` that prints the multiplication table of the given number.

Solution:

```
function printTable(num) {  
  for (let i = 1; i <= 10; i++) {  
    console.log(`${num} x ${i} = ${num * i}`);  
  }  
}
```

```
printTable(7);
```

19. Sum from 1 to N

Create a function `sumNumbers(n)` that calculates and returns the sum of numbers from 1 to n.

Example: Input: 5 | Output: 15

Solution:

```
function sumNumbers(n) {  
  let sum = 0;  
  
  for (let i = 1; i <= n; i++) {  
    sum += i;  
  }  
  
  return sum;  
}  
  
console.log(sumNumbers(5));
```